

Bayesian Classification

(Naïve Bayesian Classifier)

- A rational approach to classifying a certain set of attributes into one of the classes given some training
- Generally, used to compare the performance of a cognitive model (example a prototype, exemplar, or IBL model etc.)
- The idea is to calculate the following probability:
 - $P(\text{Belonging to a certain Class} \mid \text{Given a set of attributes})$ (E.g., $P(\text{Cancer} \mid \text{a set of brightness and relative diameter})$)
- Difference between $P(\text{Belonging to a certain Class} \mid \text{Given a set of attributes})$ from a Bayesian Classification approach and some other cognitive model:
 - The Bayesian classification is what best is “rationally (or mathematically)” possible given the set of attributes and training given in the model

Decision Task: Classification as Student or Professor (Training Set)

NAME of training pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
T1	Gabby	Baked	Clogs	Student
T2	Gabby	Roasted	Sandals	Professor
T3	Gabby	Baked	Sandals	Student
T4	Quiet	Fried	Sandals	Professor
T5	Gabby	Fried	Clogs	Student
T6	Quiet	Baked	Sandals	Student
T7	Gabby	Fried	Sandals	Professor
T8	Quiet	Fried	Clogs	Student

FIGURE 1.1 Training set for recognizing professors and students. The possible attribute values are 'gabby' and 'quiet' for HABIT; 'baked', 'fried', and 'roasted' for EATS; and 'clogs' and 'sandals' for FOOTWEAR.

Decision Task: Classification as Student or Professor (Test/Generalization Set)

NAME of recall pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
R1	Quiet	Baked	Clogs	
R2	Quiet	Roasted	Sandals	
R3	Gabby	Roasted	Clogs	
R4	Quiet	Roasted	Clogs	

FIGURE 1.2 Recall set for recognizing professors and students. The training set is given in Fig. 1.1.

Some Terminology

- $\bar{A}$ be an array of $M \geq 1$ attributes $A_1, A_2, \dots, A_M$ for the patterns of a training set, and
- $P(\bar{A})$ be the probability that a training pattern has attribute array $\bar{A}$, regardless of the class to which the pattern belongs, the attributes having discrete values.

Suppose the training set has patterns from $m \geq 1$ classes $C_1, C_2, \dots, C_m$. For $1 \leq k \leq m$, we define the following probabilities.

- $P(C_k)$ is the probability that a training pattern belongs to class C_k . It is also known as the *prior* probability of class C_k .
- $P(C_k|\bar{A})$ is the probability that a training pattern with attribute array $\bar{A}$ belongs to class C_k . This is also known as the *posterior* probability of C_k for a given $\bar{A}$. The attributes have discrete values.
- $P(\bar{A}|C_k)$ is the *conditional* probability that a training pattern of class C_k has attribute array $\bar{A}$, the attributes having discrete values.
- $p(\bar{A}|C_k)$ is the conditional probability density that a training pattern of class C_k has attribute array $\bar{A}$, the attributes having continuous values.

P(Class | Attributes)

$$P(C_k|\bar{A}) = \frac{P(C_k)P(\bar{A}|C_k)}{P(\bar{A})}$$

$$P(C_k)P(\bar{A}|C_k)$$

$$P(\bar{A}|C_k) = \prod_{i=1}^M P(A_i|C_k)$$

<- Because of this assumption
the classifier is called “Naïve”
Bayes

$$P(C_k) \prod_{i=1}^M P(A_i|C_k)$$

Probability of belonging to a particular class $P(C_k)$

Suppose that the number of patterns in class C_k is $|C_k|$, for $1 \leq k \leq m$. Then under the conventional estimation, known as the *maximum-likelihood* estimation,

$$P(C_k) = \frac{|C_k|}{\sum_{j=1}^m |C_j|} \quad (5.3)$$

Under an alternative estimation, called the *Bayesian* estimation,

$$P(C_k) = \frac{|C_k| + 1}{m + \sum_{j=1}^m |C_j|} \quad (5.4)$$

Example for $P(C_k)$

As an example, the professor–student training set given in Fig. 1.1 has 3 patterns belonging to the class professor, and 5 to the class student. There being two classes, the value of m is 2. Hence, from Eqn (5.3), the maximum-likelihood estimate of $P(\text{professor})$ is $3/(3 + 5)$, that is $3/8$; and the Bayesian estimate of $P(\text{professor})$, from Eqn (5.4), is $(3 + 1)/(2 + 3 + 5)$, that is $4/10$, which may be simplified to $2/5$.

Next to calculate $P(A_i|C_k)$

To maximize $P(C_k) \prod_{i=1}^M P(A_i|C_k)$ of Eqn (5.2), the other probability estimates required are those for $\prod_{i=1}^M P(A_i|C_k)$. Let

- the possible values for attribute A_i be $v_{i_1}, v_{i_2}, \dots, v_{i_n}$, for $1 \leq i \leq M$ and
- $|C_k^{ij}|$ be the number of training patterns of class C_k for which the value of attribute A_i is v_j , for $1 \leq k \leq m$, for $1 \leq i \leq M$, and for $i_1 \leq j \leq i_n$.

Then, for $1 \leq k \leq m$, for $1 \leq i \leq M$, and for $i_1 \leq j \leq i_n$, according to the maximum-likelihood estimation

$$P[(A_i = v_j)|C_k] = \frac{|C_k^{ij}|}{|C_k|} \quad (5.5)$$

and according to the Bayesian estimation

$$P[(A_i = v_j)|C_k] = \frac{|C_k^{ij}| + 1}{i_n + |C_k|} \quad (5.6)$$

Example of $P(A_i | C_k)$

To illustrate, attribute EATS of the training set given in Fig. 1.1 has 3 possible values (baked, roasted, and fried). Of the 5 student patterns, none has the value of EATS = roasted. Therefore, from Eqn (5.5), the maximum-likelihood estimate of $P[(\text{EATS} = \text{roasted}) | \text{student}]$ is $0/5$, and the Bayesian estimate, from Eqn (5.6), is $(0 + 1)/(3 + 5)$, which becomes $1/8$.

All $P(A_i|C_k)$

Probability	Estimates	
	Maximum-likelihood	Bayesian
$P(\text{professor})$	3/8	4/10
$P[(\text{HABIT} = \text{gabby}) \text{professor}]$	2/3	3/5
$P[(\text{HABIT} = \text{quiet}) \text{professor}]$	1/3	2/5
$P[(\text{EATS} = \text{baked}) \text{professor}]$	0/3	1/6
$P[(\text{EATS} = \text{fried}) \text{professor}]$	2/3	3/6
$P[(\text{EATS} = \text{roasted}) \text{professor}]$	1/3	2/6
$P[(\text{FOOTWEAR} = \text{clogs}) \text{professor}]$	0/3	1/5
$P[(\text{FOOTWEAR} = \text{sandals}) \text{professor}]$	3/3	4/5
$P(\text{student})$	5/8	6/10
$P[(\text{HABIT} = \text{gabby}) \text{student}]$	3/5	4/7
$P[(\text{HABIT} = \text{quiet}) \text{student}]$	2/5	3/7
$P[(\text{EATS} = \text{baked}) \text{student}]$	3/5	4/8
$P[(\text{EATS} = \text{fried}) \text{student}]$	2/5	3/8
$P[(\text{EATS} = \text{roasted}) \text{student}]$	0/5	1/8
$P[(\text{FOOTWEAR} = \text{clogs}) \text{student}]$	3/5	4/7
$P[(\text{FOOTWEAR} = \text{sandals}) \text{student}]$	2/5	3/7

FIGURE 5.1 Maximum-likelihood and Bayesian estimates of prior and conditional probabilities from the professor-student training set given in Fig. 1.1. A few of the maximum-likelihood estimates are zero but none of the Bayesian estimates are so. The maximum-likelihood estimates of the prior probabilities $P(\text{professor})$ and $P(\text{student})$ are obtained from Eqn (5.3), and their Bayesian estimates from Eqn (5.4). The remaining are conditional probabilities: their maximum-likelihood estimates are obtained from Eqn (5.5), and the Bayesian estimates from Eqn (5.6).

Decision Task: Classification as Student or Professor (Test/Generalization Set)

NAME of recall pattern	Attributes			Class
	HABIT	EATS	FOOTWEAR	
R1	Quiet	Baked	Clogs	
R2	Quiet	Roasted	Sandals	
R3	Gabby	Roasted	Clogs	
R4	Quiet	Roasted	Clogs	

FIGURE 1.2 Recall set for recognizing professors and students. The training set is given in Fig. 1.1.

Classification of Generalization Pattern (R3)

$P(\text{professor}) \times P[(\text{HABIT} = \text{gabby}) \mid \text{professor}]$
 $\times P[(\text{EATS} = \text{roasted}) \mid \text{professor}] \times P[(\text{FOOTWEAR} = \text{clogs}) \mid \text{professor}]$
is $2/5 \times 3/5 \times 1/3 \times 1/5 = 0.016$, and the value of

$P(\text{student}) \times P[(\text{HABIT} = \text{gabby}) \mid \text{student}]$
 $\times P[(\text{EATS} = \text{roasted}) \mid \text{student}] \times P[(\text{FOOTWEAR} = \text{clogs}) \mid \text{student}]$
is $3/5 \times 4/7 \times 1/8 \times 4/7 = 0.0245$. Since the value 0.0245 obtained for student is more than 0.016 obtained for professor, recall pattern R3 is classified as student.

Classification of all Patterns

NAME of recall pattern	Attributes			CLASSIFICATION
	HABIT	EATS	FOOTWEAR	
R1	Quiet	Baked	Clogs	Student
R2	Quiet	Roasted	Sandals	Professor
R3	Gabby	Roasted	Clogs	Student
R4	Quiet	Roasted	Clogs	Student

FIGURE 5.2 Classifications of the professor-student recall patterns given in Fig. 1.2, using the Bayesian estimates of probabilities in the Naive Bayes classifier. The Bayesian estimates were taken from Fig. 5.1, and the Naive Bayes classifier of Eqn (5.2) was used.



Questions or Comments?